

Larks, Owls, and the Rift: Circadian Rhythms in Global *League of Legends* Play

A within-subject, across-server analysis of ~8,000 high-ranked players on 8 regional servers

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Abstract

Human cognition and physical performance vary across the day, and individuals differ systematically in *when* they peak — the morning-preferring “larks” and evening-preferring “owls” of chronotype research. We asked whether such rhythms are detectable in a large, naturalistic behavioural record: ranked *League of Legends* matches. Using ~92.7 million games drawn from the top ~1,000 players on each of 8 regional servers, we built per-player hourly time series of in-game behaviour (two principal components of performance, PC1 and PC2) and of a skill-outcome measure (the change in matchmaking rating, ΔMMR). Within-subject Lomb–Scargle periodograms revealed a sharp circadian peak: across servers the N-weighted best period was 23.96 h for PC1 and 23.94 h for PC2 (both $\text{SEM} \leq 0.03$ h), and the population playtime periodogram peaked at exactly 24.0 h on every server. Per-player peak phases were individually significant (Benjamini–Hochberg FDR < 0.05) for 21.4% of players on PC1 and 18.5% on PC2. Pooling these significant phases, a two-component von Mises mixture was preferred over a single component by BIC for both PC1 ($\Delta\text{BIC} = 17.6$; component peaks at 21.5 h and 7.2 h) and PC2 ($\Delta\text{BIC} = 28.4$; peaks at 9.4 h and 19.9 h) — a quantitative signature of two chronotype-like clusters. In contrast, ΔMMR carried essentially no circadian signal (0.3% of players FDR-significant; no server reached the threshold for a modality test). Behaviour, then, is strongly time-of-day structured and bimodally so, consistent with larks and owls; competitive *success* is not. We discuss the interpretation and limits of inferring chronotype from public match telemetry.

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1 Introduction

Almost every measured aspect of human physiology and behaviour oscillates with a period of approximately 24 hours, driven by an endogenous circadian clock that is entrained to the local light–dark cycle (Roenneberg et al. 2007). These rhythms reach into cognition: attention, working memory, reaction time and executive control all show reliable time-of-day modulation (Schmidt et al. 2007). Critically, the *phase* of these rhythms differs between people. The morningness–eveningness dimension — colloquially “larks” versus “owls” — is a stable, partly heritable trait (Horne and Östberg 1976; Roenneberg, Wirz-Justice, and Merrow 2003), and the distribution of chronotypes in a population is broad and roughly continuous (Adan et al. 2012).

Chronotype matters for performance. When the timing of a demanding task is misaligned with a person’s internal clock — *social jet lag* — outcomes suffer. Analysing 3.4 million learning-management-system logins, Smarr and Schirmer (2018) showed that the majority of students are chronically misaligned with their class schedules and that greater misalignment predicts lower grades. In elite sport, Facer-Childs and Brandstaetter (2015) found that an athlete’s peak performance time tracks their circadian phenotype, with intra-day swings as large as ~26%. Within gaming specifically, eveningness is over-represented and linked to disrupted sleep (Hsu et al. 2024), and competitive performance in esports is a structured, trainable skill set rather than mere reflex (Nagorsky and Wiemeyer 2020).

Online competitive gaming offers an unusually clean natural experiment for chronobiology at scale. Millions of timestamped, standardised, skill-graded matches are played around the clock, across the globe, by the same individuals over months. If circadian structure shapes human behaviour, it should be legible in this telemetry. *League of Legends* (LoL), a five-versus-five multiplayer game operated on geographically separated regional servers, is well suited to the question: each server has a known local time zone, allowing matches to be aligned to *local* hour, and matchmaking continuously estimates each player’s skill.

We test three questions:

1. **Existence.** Do individual players show ~24 h rhythms in their in-game behaviour?
2. **Bimodality (larks vs owls).** If rhythms exist, is the population of peak times *bimodal* — clustering into morning-peaking and evening-peaking groups — rather than a single mode?
3. **Consequence.** Does the *outcome* of play (skill change, ΔMMR) carry the same circadian signature as behaviour, or is success time-invariant?

We address these with a within-subject spectral analysis of ~8,000 high-ranked players, replicated independently across 8 servers and combined by an N-weighted meta-analysis.

2 Methods

2.1 Data provenance and aggregation

The analysis reuses the large longitudinal *League of Legends* cohort provided by Riot Games and described by Aung et al. (Aung et al. 2018) and Vardal et al. (Vardal et al. 2022). Its defining feature is the sampling design: a random sample of players who registered **new accounts at the start of a competitive season** — accounts with no prior ranked history — followed across **one entire season** (21 January to November 2016) of default five-versus-five ranked Solo/Duo-queue play. Because every account enters ranked play at the same default matchmaking rating and is then observed continuously, the cohort offers an unusually clean within-subject view of each player’s full activity and skill trajectory — the property our within-player periodogram and ΔMMR analyses depend on. Each match row carries the per-minute performance fields we reduce by PCA together with the player’s pre-match rating, from which ΔMMR is derived. The same data lineage underpins prior work on this cohort: early-season learning rate predicts end-of-season skill (Aung et al. 2018), and the temporal *spacing* of practice (rather than its timing) shapes performance gains (Vardal et al. 2022).

Raw match records were supplied by Riot Games as a columnar Parquet store and queried through an embedded DuckDB analytical database (Raasveldt and Mühleisen 2019), from which an hourly aggregate table was materialised. We analysed the **eight regional servers** used by the source-dataset papers (Aung et al. 2018; Vardal et al. 2022) — BR1, EUN1, EUW1, JP1, LA1, LA2, NA1 and OC1 — together contributing ~92.7 million ranked games; the remaining platforms (ID1, TR1 and the PBE1 public-beta realm) were excluded as small or unreliable. For each server we selected the `top_n_players = 1000`

most active accounts and retained the first `max_hour_limit` = 5000 hours (~208 days) of each player’s history to bound edge effects. Each match timestamp was converted to the server’s **local hour** using a fixed per-server UTC offset, so that “9 a.m.” means the same wall-clock experience on every server.

2.2 Behavioural features and principal components

Seven rate features were derived per game — gold, damage dealt, kills, deaths, assists, neutral creeps and enemy creeps, each per minute — plus the fraction of time spent dead. Population-level hourly features were standardised and reduced by sign-oriented principal component analysis (SVD); per-game player records were then projected onto this fixed basis so that every player is scored on a common set of axes (Harris et al. 2020). The first two components dominate:

- **PC1** ($\approx 61\%$ of variance) loads *positively* on all activity features — a general **in-game output / engagement intensity** axis (high PC1 = a busy, high-tempo game).
- **PC2** ($\approx 20\%$ of variance) contrasts farming/economy (creeps, gold) against time spent dead — an **economy-versus-attribution** axis.

A complementary “success-aware” PCA that included win rate was computed for descriptive comparison. As a direct skill-outcome metric we also computed ΔMMR , the within-player change in matchmaking rating from one game to the next.

2.3 Within-subject periodograms

For each player and each metric (PC1, PC2, ΔMMR) we placed the game-level scores on their irregular real-time stamps and computed a Lomb–Scargle periodogram (Lomb 1976; Scargle 1982) over periods of 6–48 h (`player_n_freq` = 2000 frequencies) using `astropy.timeseries.LombScargle` (Astropy Collaboration 2018; VanderPlas 2018). The Lomb–Scargle estimator is appropriate here because match times are unevenly spaced. The (evenly spaced) frequency grid is **anchored so that exactly 24 h is a grid node**, allowing a true circadian peak to be reported at 24.0 h rather than snapping to a neighbouring bin. Per-player periodograms were then **incoherently averaged** within each server–metric cell (averaging *power*, which is phase-blind), and the period of maximum mean power was recorded as that cell’s best period. A population-level periodogram of raw activity (`TIMEPLAYED`) and of win rate was computed analogously over 6–48 h.

2.4 Phase estimation and multiple-testing control

To locate *when* in the day each player peaks, we fitted a fixed 24 h sinusoid by ordinary least squares to each player’s series, converted the fitted lag to a **local peak hour** in $[0, 24)$, and tested the joint significance of the sinusoid. Across the players in each server–metric cell, the resulting *p*-values were corrected with the Benjamini–Hochberg false-discovery-rate procedure (Benjamini and Hochberg 1995) at $\alpha = 0.05$. Only FDR-significant peak phases entered the modality analysis.

2.5 Circular modality: one versus two chronotypes

Peak hours are circular. For each server–metric cell with at least 20 FDR-significant phases we fitted two circular models to the phases (expressed as angles) (Fisher 1993; Mardia and Jupp 2000; Berens 2009):

- a **one-component von Mises** distribution (mean direction μ and concentration κ estimated from the resultant length); and
- a **two-component von Mises mixture** fitted by expectation–maximisation.

Models were compared by the Bayesian Information Criterion (Schwarz 1978), charging 2 free parameters to the one-component fit and 5 to the mixture; a positive $\Delta\text{BIC} = \text{BIC}_1 - \text{BIC}_2$ favours two components, i.e. two chronotype-like clusters. We report both a per-server tally and a final **pooled** fit that aggregates significant phases across servers (using only cells with ≥ 20 phases). As an assumption-light companion, we also summarised the same phases with phase-count-weighted circular kernel density estimates ($\kappa = 4$).

2.6 Across-server meta-analysis and quality control

All headline numbers are **N-weighted across servers**: period peaks are weighted by the number of valid players, phase fractions by players analysed, and population metrics by games played. Implementation

used NumPy (Harris et al. 2020), SciPy (Virtanen et al. 2020), pandas (McKinney 2010) and Matplotlib (Hunter 2007).

i Outlier handling

Two server-level period estimates — both from **JP1** — were robust outliers and were excluded from the period meta-analysis: its Δ MMR peak (44.4 h, an implausible near-double of the circadian period) and its PC1 peak (8.0 h, a flip onto the third harmonic of the daily cycle). Outliers were flagged *within* each metric by a median/MAD modified z -score ($|z| > 3.5$) combined with an absolute-deviation gate (> 3 h from the median); the gate keeps genuine near-circadian values — peaks one or two grid bins either side of 24 h — while still catching harmonics and long-period artifacts, and it also decides the case where the 24 h-anchored grid collapses the MAD to zero by placing most servers on the same bin. After exclusion PC1 and Δ MMR rest on 7 servers / 7,000 players each; PC2 retains all 8.

3 Results

3.1 Behaviour is rhythmic; success is not

Within-subject periodograms show a sharp circadian peak for behaviour and only noise for outcome. Figure 1 illustrates this for a single representative server (NA1): PC1 and PC2 each produce a clean spike at ~ 24 h above a flat background, whereas Δ MMR yields a broad, low, ragged spectrum with no dominant period.

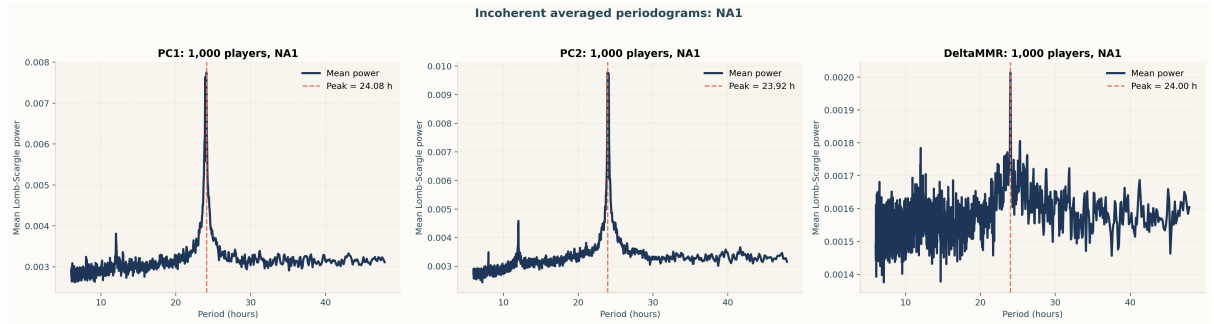


Figure 1: Incoherently averaged within-subject Lomb–Scargle periodograms for one server (NA1, 1,000 players). PC1 (left, peak 24.08 h) and PC2 (centre, 23.92 h) show sharp circadian spikes above a flat background; Δ MMR (right) is a weak, noisy spectrum with peak power roughly an order of magnitude smaller, its nominal 24.00 h peak barely rising above the noise.

This pattern is consistent across all servers. The N-weighted best period (Figure 2, left) is **23.96 h for PC1** (SEM 0.029) and **23.94 h for PC2** (SEM 0.020), both indistinguishable from 24 h, with very small between-server scatter; the Δ MMR peak averages **exactly 24.00 h** (after removing the JP1 outlier) but rests on a far weaker peak (Table 1).

Table 1: N-weighted within-subject best periods.

Metric	Servers	Players	Best period (h)	SEM (h)
PC1	7	7,000	23.96	0.029
PC2	8	8,000	23.94	0.020
Δ MMR	7	7,000	24.00	0.015

The fraction of *individually* FDR-significant players tells the same story (Figure 2, right): **21.4%** of players have a significant ~ 24 h rhythm in PC1 and **18.5%** in PC2, but only **0.3%** do in Δ MMR (26 of 8,000 players). Behaviour is strongly clock-locked; the *outcome* of competition is essentially clock-agnostic.

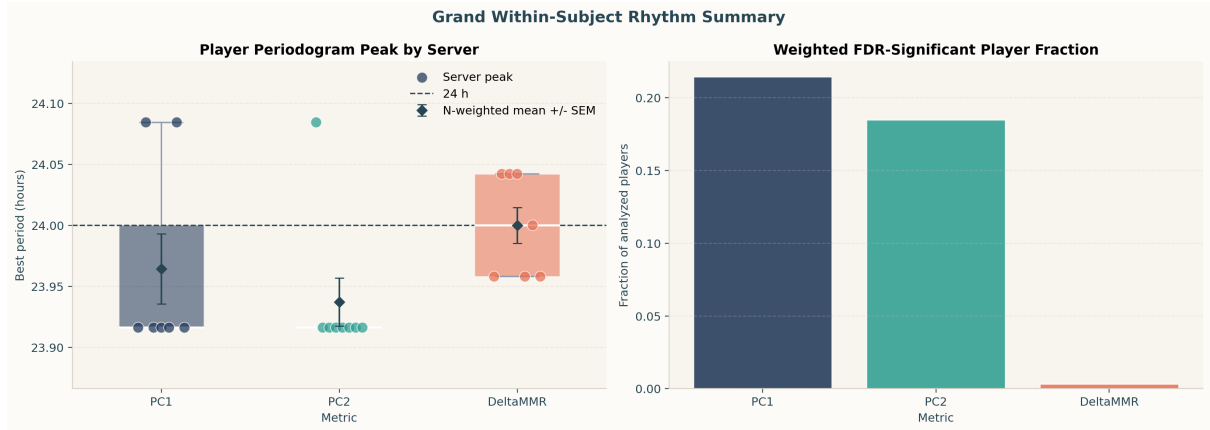
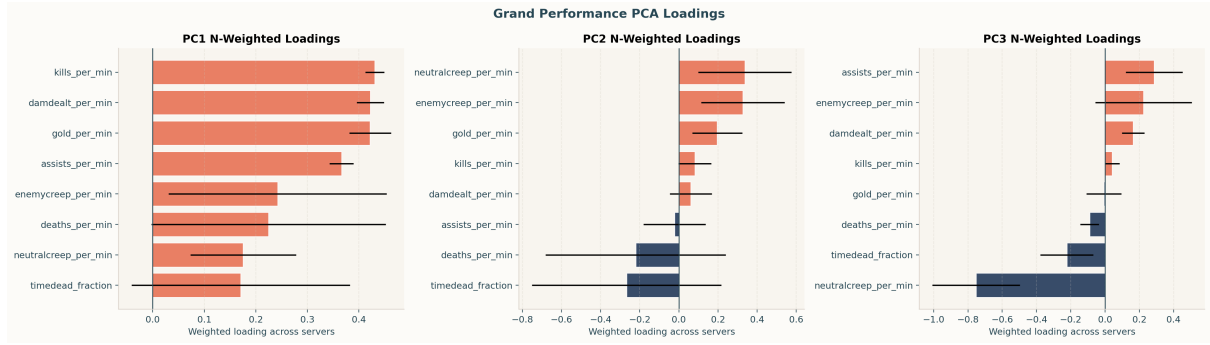


Figure 2: Within-subject rhythm summary across 8 servers. **Left:** per-server best period for each metric (dot size $\propto$ valid players) with the N-weighted mean $\pm$ SEM (diamonds) and the 24 h reference (dashed). After outlier screening (JP1 removed from PC1 and Δ MMR), all three metrics sit at $\sim$ 24 h. **Right:** N-weighted fraction of players whose peak phase is FDR-significant — high for the behavioural PCs, near zero for Δ MMR.

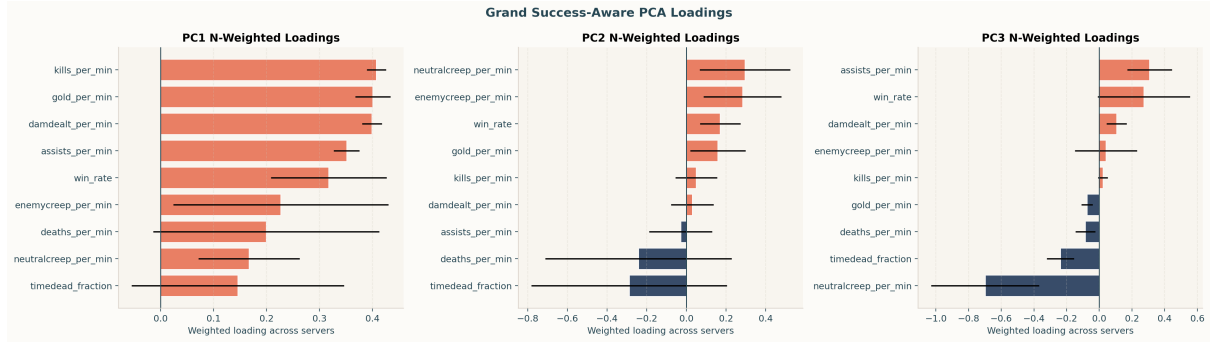
For reference, the population-level periodograms behave as expected: with the 24 h-anchored grid the raw activity periodogram (TIMEPLAYED) peaks at **exactly 24.0 h on every one of the eight servers**, and raw win rate peaks near 24 h on average (N-weighted 23.6 h, though noisier across servers).

3.2 What the components mean

Figure 3 shows the N-weighted PCA loadings. PC1 loads positively on every activity feature, confirming it as an overall output/engagement axis; PC2 separates farming and economy from time spent dead. The success-aware decomposition (lower panel) places only a modest positive win-rate loading on its leading component (N-weighted 0.29), reinforcing that the dominant behavioural axes are about *how* a game is played, not whether it is won.



(a) Performance PCA: PC1 is an all-positive output/engagement axis (~62% variance); PC2 contrasts economy against time-dead (~19%).



(b) Success-aware PCA (win rate included): win rate loads only modestly on the leading components.

Figure 3: N-weighted PCA loadings across servers; whiskers show between-server spread.

3.3 Two chronotypes: the peak-phase distribution is bimodal

The central question is the *shape* of the peak-time distribution. Pooling all FDR-significant phases and comparing circular models, **both behavioural components prefer a two-component von Mises mixture** (Figure 4, Table 2):

- **PC1:** $\Delta\text{BIC} = 17.6$ in favour of two components, with peaks at **21.5 h** and **7.2 h** (weights 0.57 / 0.43) — an evening-peaking “owl” cluster and a morning-peaking “lark” cluster of nearly equal size.
- **PC2:** $\Delta\text{BIC} = 28.4$, with peaks at **9.4 h** and **19.9 h** (weights 0.65 / 0.35) — again a morning and an evening mode.

ΔMMR could not be tested for modality at all: no server reached the 20-significant-phase threshold, so the pooled ΔMMR cell is empty.

Table 2: Pooled circular bimodality test on FDR-significant peak phases.

Metric	Servers	Phases	Preferred	ΔBIC (1–2)	Peak 1 (h)	Peak 2 (h)	Weights
PC1	8	1,715	2- component	17.6	21.5	7.2	0.57 / 0.43
PC2	7	1,465	2- component	28.4	9.4	19.9	0.65 / 0.35
ΔMMR	0	0	—	—	—	—	—

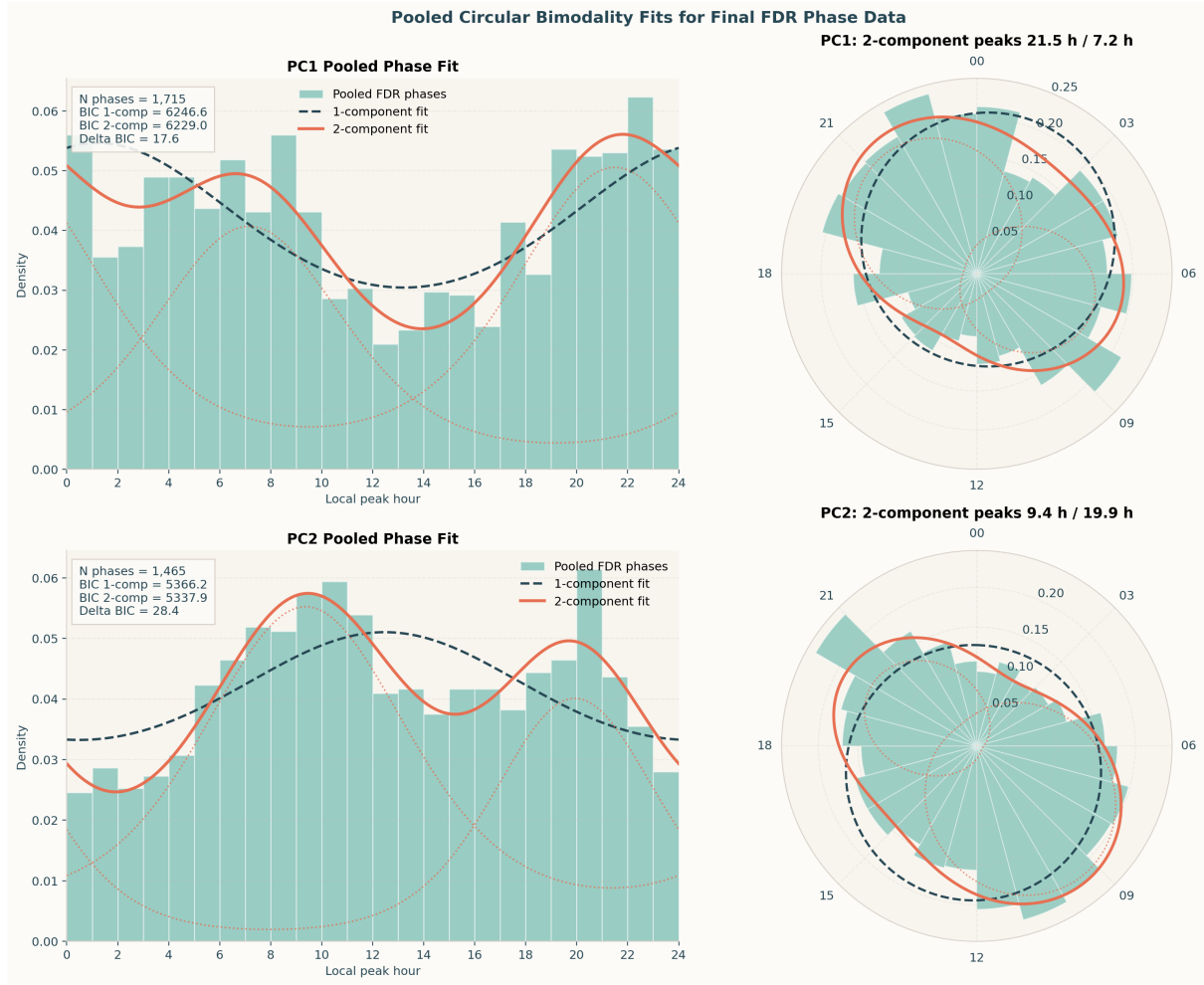


Figure 4: Pooled circular bimodality fits. For PC1 (top) and PC2 (bottom): histograms of FDR-significant local peak hours with the one-component fit (dashed) and two-component mixture (solid; dotted components), shown in both hour-domain (left) and polar (right) views. Both metrics show two clear modes — a morning and an evening cluster — and the mixture is preferred by BIC.

The per-server tally is consistent: across servers, FDR-significant PC1 phases split 818 (one-component-preferred) to 897 (two-component-preferred) and PC2 704 to 761 — i.e. a majority of server-level phases independently favour the two-cluster description, which is what produces a decisive *pooled* preference once power is aggregated.

An assumption-light kernel-density view (Figure 5) reproduces the same landscape: the phase-count-weighted circular density of PC1 peaks concentrates around the late evening (weighted peak 22.4 h) while PC2 peaks in the morning (9.8 h), each with a secondary lobe — the smooth-density counterpart of the two-component fits.

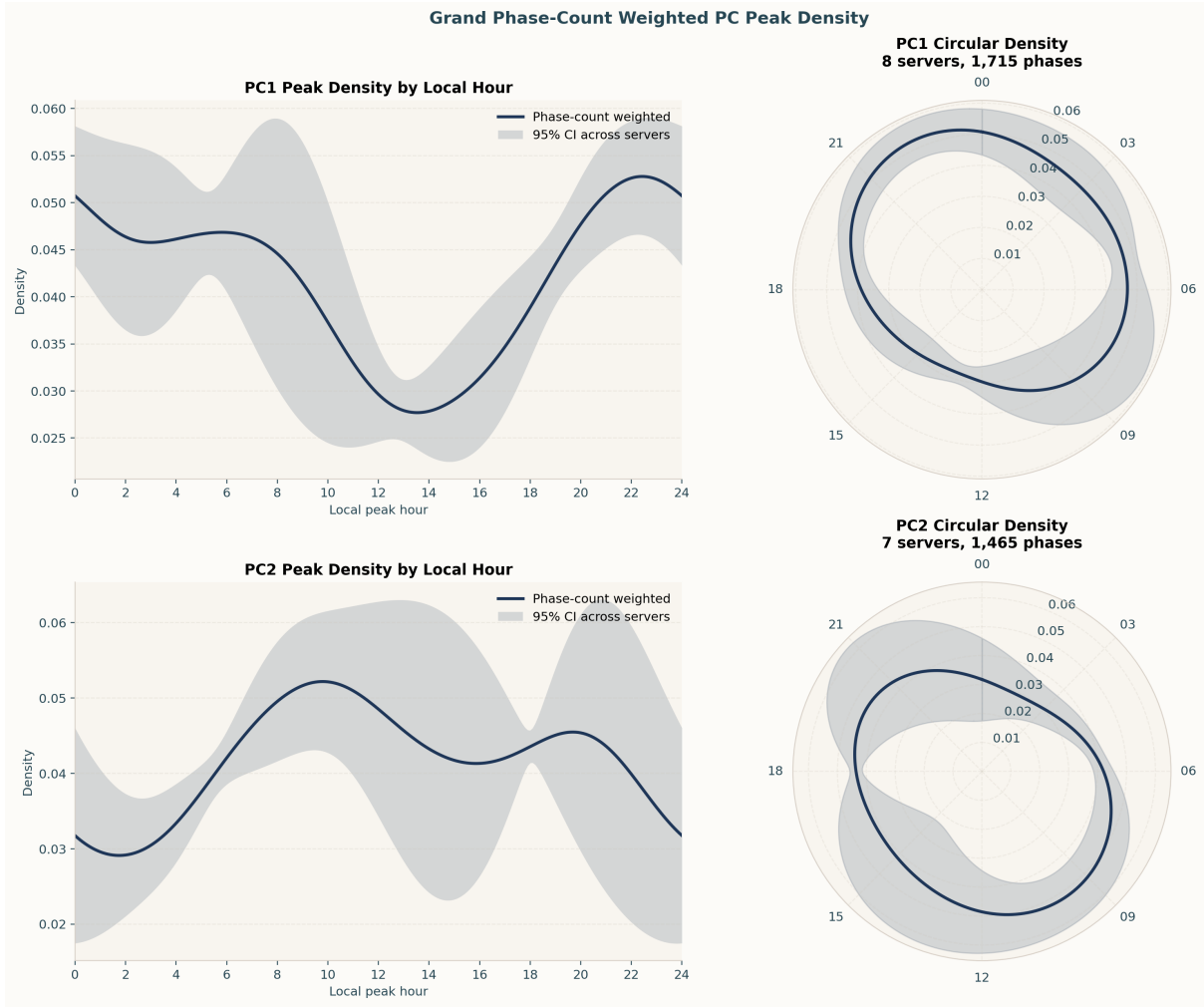


Figure 5: Phase-count-weighted circular kernel density of FDR-significant peak hours ($\kappa = 4$), with 95% across-server bands. The bimodal structure seen in the von Mises fits is reproduced without assuming a parametric mixture.

3.4 When do players actually win?

Although individual skill *change* shows no rhythm, the aggregate win rate does vary with local hour (Figure 6). Pooled across servers, win rate is highest in the early-morning hours (~03:00–06:00 local) and lowest through the afternoon and early evening. The absolute swing is small — raw win rate moves only between about 0.483 and 0.491 — and, because this is computed over *whichever* players are online at each hour, it reflects a mixture of who plays when (composition) and any genuine time-of-day effect, not a within-person performance curve. We therefore treat it as descriptive context rather than a chronotype result.

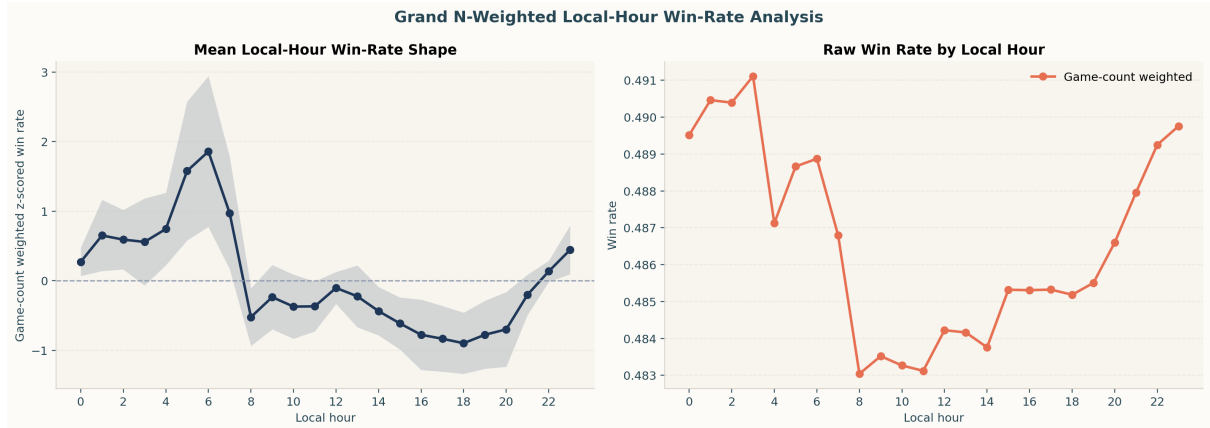


Figure 6: N-weighted win rate by local hour. **Left:** z-scored win-rate shape with across-server band, peaking in the early morning. **Right:** raw game-count-weighted win rate (note the narrow 0.483–0.491 range).

4 Discussion

Three findings stand out. First, **circadian rhythms are real and strong in LoL play behaviour**: within-subject periodograms peak crisply at 24 h for both behavioural principal components, the estimate is stable to three significant figures across 8 independent servers, and more than one in five players carries an individually significant daily rhythm. This is, to our knowledge, one of the larger naturalistic demonstrations of behavioural circadian structure outside the laboratory, and it complements educational (Smarr and Schirmer 2018) and athletic (Facer-Childs and Brandstaetter 2015) big-data work.

Second, **the population of peak times is bimodal**. Both PC1 and PC2 prefer a two-component von Mises mixture, recovering a morning cluster and an evening cluster in each case. This is exactly the structure predicted if the player base is a mixture of larks and owls (Horne and Östberg 1976; Roenneberg, Wirz-Justice, and Merrow 2003), and the result survives a stringent pipeline (per-player FDR control, a BIC penalty against the more complex model, and an outlier-screened meta-analysis) as well as a non-parametric density cross-check. We are careful about the label: what we measure is a *behavioural* phase — the hour at which a player’s in-game output peaks — not a physiological dim-light melatonin onset. Behavioural phase is shaped by chronotype but also by school, work and social schedules. The bimodality is genuine; equating each mode one-to-one with a biological chronotype is an interpretation, not a measurement.

Third, **competitive success is essentially time-invariant**. Δ MMR shows no within-subject circadian rhythm (0.3% FDR-significant; no server testable for modality), even though the very same players, on the very same games, show clear rhythms in *how* they play. The matchmaking system is designed to hold win rate near 50% by pairing players of similar skill, which mechanically suppresses any outcome rhythm; the small aggregate win-rate variation by hour (Section 3.4) is better explained by *who* is online than by a within-person performance curve. The contrast with timescale is instructive: on this very cohort, *across*-season skill change is highly structured and predictable — early-season learning rate forecasts end-of-season rating (Aung et al. 2018) — yet the *within*-day timing of skill change leaves no detectable circadian mark. Indeed, prior analyses of this same dataset deliberately set MMR aside as a match-to-match performance measure, precisely because it is governed by the (opaque, win/loss-weighted) matchmaking algorithm rather than by momentary individual play (Vardal et al. 2022); our null is the chronobiological counterpart of that caution. The practical reading for a competitive player is encouraging and slightly deflating at once: your *style* swings with the clock, but the ladder largely launders away the consequences.

4.1 Limitations

This is an observational analysis of a self-selected, highly active, high-ranked population (top ~1,000 per server), so the chronotype mixture we recover need not match the general public. Phases are inferred from a fitted 24 h sinusoid on irregular match times rather than from continuous physiological recording.

Local hour uses a single per-server UTC offset and does not model daylight-saving transitions or the within-region spread of true time zones. The two-component von Mises model is the simplest test for bimodality and does not exclude richer multi-modal or continuous structure. Finally, the headline meta-analysis weights servers by N ; alternative weightings (e.g. equal-server) give qualitatively the same conclusions but slightly different point estimates.

4.2 Conclusion

Across 8 global servers and ~8,000 players, *League of Legends* behaviour is clearly circadian and the distribution of personal peak times is bimodal — a data-driven echo of larks and owls — while competitive outcome is not. The match telemetry of a popular game turns out to be a surprisingly sharp instrument for human chronobiology, and the analysis pipeline here (within-subject Lomb–Scargle, FDR-controlled phase estimation, and BIC-compared circular mixtures, meta-analysed across servers) is a reusable template for similar questions in other large behavioural datasets.

Data and code availability

Analysis code (`riot_analysis.py`, `grand_analysis.py`) and the final per-server and grand outputs are in the project repository. The raw Riot Parquet store is not redistributed; the pipeline rebuilds a local DuckDB cache from it on demand. All figures in this paper are generated by the grand analysis and live under `results/GRAND/` and `results/<SERVER>/`.

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